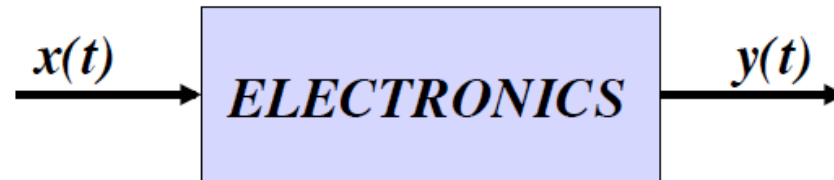


دیجیتال کردن سیگنال گفتار

Continuous time vs Discrete time

◆ Continuous time system

- Good for analogue & general understanding
- Appropriate mostly to analogue electronic systems



◆ Electronics are increasingly digital

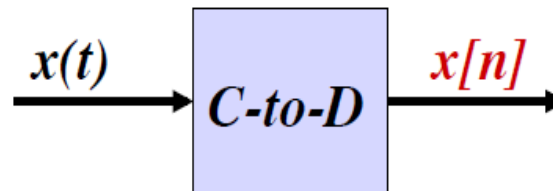
- E.g. mobile phones are all digital, TV broadcast is will be 100% digital in UK
- We use digital ASIC chips, FPGAs and microprocessors to implement systems and to process signals
- Signals are converted to numbers, processed, and converted back



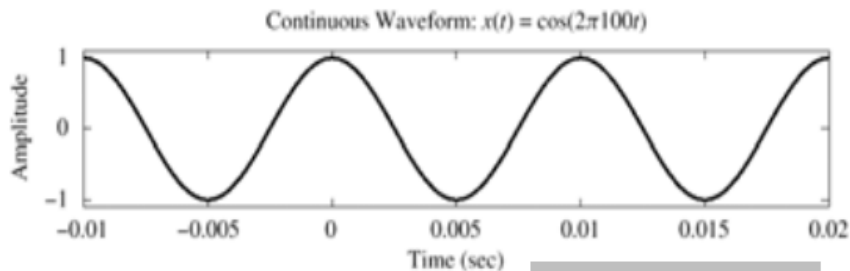
Sampling Process

- ◆ Use A-to-D converters to turn $x(t)$ into numbers $x[n]$
- ◆ Take a sample every sampling period T_s – uniform sampling

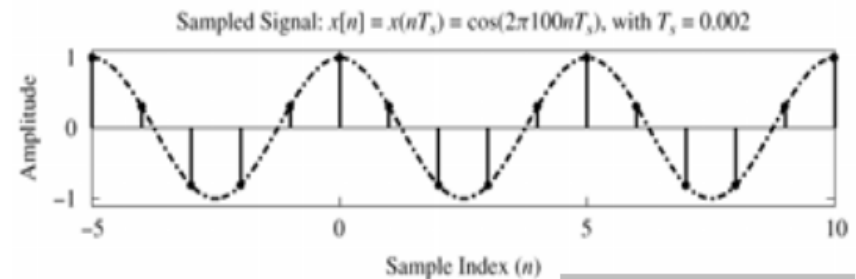
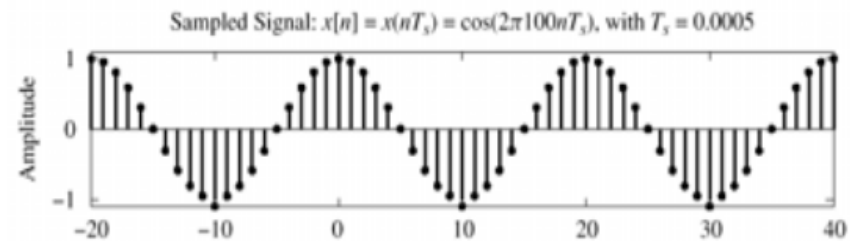
$$x[n] = x(nT_s)$$



$$f_s = 2 \text{ kHz}$$



$$f = 100\text{Hz}$$



$$f_s = 500\text{Hz}$$

Sampling Theorem

- ◆ Bridge between continuous-time and discrete-time
- ◆ Tell us HOW OFTEN WE MUST SAMPLE in order not to lose any information

Sampling Theorem

A continuous-time signal $x(t)$ with frequencies no higher than $f_{\max}$ (Hz) can be reconstructed EXACTLY from its samples $x[n] = x(nT_s)$, if the samples are taken at a rate $f_s = 1/T_s$ that is greater than $2f_{\max}$.

- ◆ For example, the sinewave on previous slide is 100 Hz. We need to sample this at higher than 200 Hz (i.e. 200 samples per second) in order NOT to lose any data, i.e. to be able to reconstruct the 100 Hz sinewave exactly.
- ◆ $f_{\max}$ refers to the maximum frequency component in the signal that has **significant** energy.
- ◆ Consequence of violating sampling theorem is **corruption of the signal** in digital form.

Saving Sampled Speech Signal

Discrete Time	Real Numbers	
t= 1	X(1)	Round To Integers (Quantization) How many bits are required to save these integers numbers? Usually 16 bit (Bit Per Sample)
t= 2	X(2)	
t= 3	X(3)	
...		
...		
t= n	X(n)	

Saving Sampled Speech Signal-Wav file Format

- ▶ Wav file size : 44 byte header + samples size
- ▶ Example: A speech wav file contains 6 seconds speech with sampling frequency 16 khz where each sample is saved by 16 bits. What is wavfile size in byte?
- ▶ $6 \times 16000 = 96000$ sample per speech file
- ▶ Bits required: 96000×16
- ▶ Byte required: $96000 \times 2 = 192000$ byte
- ▶ Wavfile size = $44 + 192000 = 192044$ byte ≈ 192.044 kB

MATLAB Function For Reading Wav File

- ▶ Wavread
- ▶ In new matlab versions: **auread**
 - ▶ `[x,Fs,Bits] = wavread('wav file path and name')`
 - ▶ x: speech signal
 - ▶ Fs: Sampling Frequency
 - ▶ Bits : bit per samples
- ▶ Wavwrite for writing wavfile